

A Cloud-Based Google Earth Engine Application for Multi-Scale Geographic Accessibility Analysis to Healthcare Facilities in Peru

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Abstract

Geographic accessibility is a fundamental determinant of health equity and universal health coverage, particularly in countries with complex topography such as Peru. This study develops a Google Earth Engine-based application to assess multi-scale healthcare accessibility and spatial inequalities nationwide. Healthcare facility data from RENIPRESS were integrated with satellite-derived elevation, population, travel time, and nighttime radiance datasets, and accessibility was evaluated using buffer distances ranging from 5 km to 20 km while accounting for terrain, isolation, and socioeconomic context. Results show a strong concentration of healthcare facilities in Lima and major urban centers, whereas rural and highland areas experience substantial geographic barriers. Basic health posts dominate the system, while hospitals are scarce and unevenly distributed. The proposed cloud-based framework enables reproducible national accessibility assessments and supports evidence-based planning to reduce geographic health inequalities in Peru.

Keywords: Healthcare accessibility; GIS; Google Earth Engine; Universal health coverage; Peru.

1 Introduction

Geographic accessibility to healthcare facilities is a fundamental determinant of health equity and a critical component of achieving universal health coverage (UHC), as emphasized by the World Health Organization (WHO) and Sustainable Development Goal 3.8 [1, 2]. The WHO recommends that populations reside within 5 km of a healthcare facility to ensure reasonable access to essential health services [3]. However, this standard presents significant challenges in geographically complex nations like Peru, where extreme topographic variation—ranging from coastal deserts at sea level to Andean peaks exceeding 6,000 meters and dense Amazonian rainforests—creates substantial barriers to healthcare delivery [4, 5]. These geographic constraints, combined with highly dispersed rural populations, necessitate adapted accessibility metrics that account for local terrain and infrastructure realities [6].

Traditional methods for assessing healthcare accessibility typically rely on Geographic Information Systems (GIS) that require substantial computational resources, specialized software licenses, and technical expertise in geospatial analysis [7, 8]. These barriers limit the capacity of many health ministries, regional governments, and research institutions—particularly in low- and middle-income countries (LMICs)—to conduct routine accessibility assessments and monitor temporal trends [9, 10]. Moreover, conventional approaches often involve labor-intensive workflows including manual data acquisition, pre-processing, geometric corrections, and spatial analyses that can take weeks to complete for national-scale assessments [11].

The advent of cloud-based geospatial platforms, particularly Google Earth Engine (GEE), has revolutionized accessibility by providing free access to petabyte-scale satellite imagery archives and distributed computing infrastructure [12, 13]. Recent applications of GEE in public health have demonstrated its potential for large-scale environmental health monitoring, disease mapping, and infrastructure planning [14–16].

Peru’s healthcare system, organized through the National Superintendency of Health (SUSALUD), maintains the National Registry of Health Establishment and Services (RENIPRESS), which catalogs over 8,000 public and private facilities categorized by their complexity and service capacity [17]. Despite this comprehensive registry, systematic spatial accessibility assessments remain limited, with most studies focusing on specific regions or diseases rather than providing national-scale coverage analysis [18, 19]. Previous research has identified significant urban-rural disparities in healthcare access, with rural populations experiencing 2-3 times longer travel times to facilities compared to urban residents [20]. Furthermore, the private health sector, which accounts for approximately 30% of facilities, demonstrates marked geographic clustering in economically developed urban areas, potentially exacerbating inequities [21, 22].

Geographic barriers in Peru are not merely topographic but multidimensional, en-

compassing elevation-related physiological challenges (hypoxia above 3,000m affects both patients and healthcare workers), steep terrain limiting vehicular access (slopes exceeding 20° restrict ambulance services), and temporal isolation from urban centers (travel times exceeding 180 minutes to cities with >50,000 inhabitants) [23, 24]. These barriers differentially affect vulnerable populations, including indigenous communities in the Amazon basin and Quechua-speaking populations in the Andes, who already face linguistic and cultural barriers to healthcare utilization [25, 26]. Understanding the spatial distribution of these barriers is essential for evidence-based health policy, particularly for prioritizing new facility construction, mobile health services deployment, and telemedicine infrastructure investments [27, 28].

Recent advances in satellite remote sensing provide unprecedented opportunities to characterize these geographic and socioeconomic determinants of accessibility. The Shuttle Radar Topography Mission (SRTM) offers 30-meter resolution digital elevation models enabling precise terrain analysis [29]. WorldPop datasets provide gridded population estimates at 100-meter resolution, crucial for calculating population-weighted accessibility metrics [30]. The Oxford Malaria Atlas Project’s friction surfaces model travel time to cities based on road networks, terrain, and transportation infrastructure [31]. Additionally, nighttime radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) serves as a proxy for economic development and urbanization, allowing analysis of socioeconomic determinants of facility distribution [32, 33].

Building on methodological frameworks established in remote sensing applications for environmental monitoring [34], we developed an automated, cloud-based application using GEE to quantify healthcare accessibility in Peru through multi-scale buffer analysis (5km, 10km, 15km, 20km). Following validation approaches similar to those used in vegetation monitoring studies [35], we compare our coverage estimates against official statistics from Peru’s Demographic and Family Health Survey (ENDES) and Ministry of Health reports. Our methodology examines how institutional characteristics (facility category, sector, location), geographic barriers (elevation, slope, travel time), population distribution, and socioeconomic development (nighttime radiance) explain variations in the dependent variable: population coverage within 10km radius—an adapted threshold accounting for Peru’s challenging geography [36].

1.1 Research Objectives

This study addresses three primary research questions:

1. What proportion of Peru’s population has geographic access to healthcare facilities within 10km, and how does this compare to WHO standards?

2 Method Details

The methodology integrates user-defined parameters and automated server-side geospatial processing within the Google Earth Engine (GEE) platform. The workflow begins with the definition of the study area (Peru national boundaries), followed by automated data acquisition, buffer generation around healthcare facilities, population coverage calculation, and validation against official statistics. The analysis leverages GEE’s cloud infrastructure to process multiple satellite datasets without requiring local computational resources or manual data downloads.

2.1 Study Area and Healthcare Facility Database

2.1.1 Geographic Context

Peru encompasses 1,285,216 km² of highly heterogeneous terrain across three distinct geographic zones: the Pacific coastal plain, the Andean mountain range with peaks exceeding 6,000m, and the Amazon basin. The national population of approximately 33.7 million (2020 WorldPop estimate) is highly concentrated in the Lima metropolitan region while rural Amazonian and Andean communities remain dispersed across challenging terrain.

2.1.2 RENIPRESS Database

The primary dataset consists of the National Registry of Health Institutions and Services (RENIPRESS) maintained by Peru’s National Superintendency of Health (SUSALUD). The registry was accessed in January 2024 and contains 21,379 active healthcare facilities distributed across all 25 departments. Each facility record includes unique identification code, facility category (I-1 through III-2), institutional sector (MINSA/Regional Government, EsSalud, Private, Armed Forces/Police), administrative location (Department, Province, District), and geographic coordinates in decimal degrees (EPSG:4326).

RENIPRESS data were downloaded in CSV format and underwent quality control including removal of inactive facilities, validation of coordinate ranges to exclude erroneous locations, and conversion to GEE-compatible format with point geometries. The cleaned dataset was uploaded as a custom GEE asset for reproducible analysis.

2.2 Application Design and Buffer-Based Accessibility Analysis

2.2.1 Platform Architecture

The application was developed using the GEE JavaScript API and deployed as a web-based interface accessible through standard web browsers. The platform consists of three components: client-side user interface for parameter input, server-side GEE processing

for geospatial computations, and pre-loaded satellite imagery with uploaded healthcare facility locations.

2.2.2 Region of Interest Definition

The analytical workflow allows users to define study areas through multiple methods. For national analysis, Peru’s administrative boundaries are loaded from FAO GAUL 2015 at country and department levels. Users can select specific departments for sub-national analysis or draw custom polygons directly on the map interface.

2.2.3 Buffer Generation and Population Coverage Calculation

Geographic accessibility was quantified using euclidean buffer analysis. For each health-care facility, circular buffers were generated at four radii: 5km (WHO standard), 10km (primary analysis threshold adapted for Peru’s terrain), 15km, and 20km. The 10km threshold represents a compromise between international standards and local geographic realities, where this distance can represent one to three hours of actual travel time depending on terrain and infrastructure.

Buffers were merged using spatial union operations to eliminate overlapping areas and prevent double-counting of populations. The merged coverage area was converted to a binary mask and applied to the WorldPop population raster. Population within coverage was calculated by summing all 100m grid cells intersecting the buffer zone, constrained to Peru’s national boundary. Coverage percentage was computed as the ratio of covered population to total national population. The analysis was repeated for all four buffer distances to assess sensitivity to radius selection.

2.2.4 Economic Development Proxy

Mean nighttime radiance within 1km radius of each facility was calculated from VIIRS data to characterize local economic context. This metric captures the economic environment in which facilities operate, with higher values indicating urban and developed areas. Facilities were categorized as urban (radiance exceeding $10 \text{ nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$), semi-urban ($3\text{--}10 \text{ nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$), or rural (below $3 \text{ nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$) based on observed thresholds in the data distribution.

2.3 Validation Methodology

To assess accuracy of GEE-derived coverage estimates, results were compared against three independent sources: ENDES 2022 (Peru’s Demographic and Family Health Survey providing household-level distance data), MINSA 2023 departmental accessibility reports

from field surveys, and INEI 2023 census-based estimates for Lima metropolitan region. Agreement percentage was calculated as:

$$\text{Agreement (\%)} = \left(1 - \frac{|\text{Calculated} - \text{Reference}|}{\text{Reference}} \right) \times 100 \quad (1)$$

Values exceeding 95% indicate high accuracy consistent with validated remote sensing applications.

3 Results

3.1 National Healthcare Coverage Analysis

The analysis of 21,379 active healthcare facilities from the RENIPRESS database revealed significant insights into Peru’s geographic accessibility to health services. Using the 10km buffer threshold adapted for Peru’s challenging terrain, the national coverage analysis demonstrates the spatial distribution of healthcare access across the country’s diverse geographic zones.

Illustrates the 10km coverage zones surrounding all healthcare facilities, showing pronounced clustering along the Pacific coastal plain and inter-Andean valleys, with notably sparser coverage in the Amazonian lowlands and high-Andean regions. The concentration of facilities in Lima department (9,612 facilities, 45% of national total) reflects both population distribution patterns and historical infrastructure development priorities.

3.2 Healthcare Facility Distribution

3.2.1 Distribution by Category

The national healthcare network exhibits a pyramidal structure consistent with primary care-oriented systems, Table 1). Category I-1 facilities (basic health posts without physicians) constitute the largest proportion at 44.9% (9,594 facilities), followed by Category I-2 facilities (health posts with physicians) at 30.3% (6,480 facilities). Health centers (Categories I-3 and I-4) represent 21.6% of facilities, while hospitals (Categories II and III) account for only 3.3% (693 facilities total).

Table 1: Distribution of Healthcare Facilities by Complexity Category

Category	Description	Facilities	Percentage
I-1	Basic health post	9,594	44.9%
I-2	Health post with physician	6,480	30.3%
I-3	Health center	4,188	19.6%
I-4	Health center with beds	424	2.0%
II	General hospitals	634	3.0%
III	Specialized hospitals	59	0.3%
Total		21,379	100%

3.2.2 Distribution by Institutional Sector

The institutional sector analysis reveals the mixed public-private composition of Peru’s health system. Public sector facilities (MINSA and Regional Governments) dominate the network, particularly in rural and underserved areas where private sector presence is minimal. The private sector demonstrates marked geographic clustering in economically developed urban zones, as evidenced by the correlation between facility location and nighttime radiance values.

3.2.3 Geographic Distribution by Department

Figure 1 presents the top 10 departments by facility count, highlighting extreme concentration in Lima (9,612 facilities), followed by Piura (1,490), Cajamarca (1,411), Arequipa (1,195), and Cusco (1,187). This distribution reflects both population density patterns and regional health system capacity, with coastal and inter-Andean departments showing higher facility density than Amazonian regions.

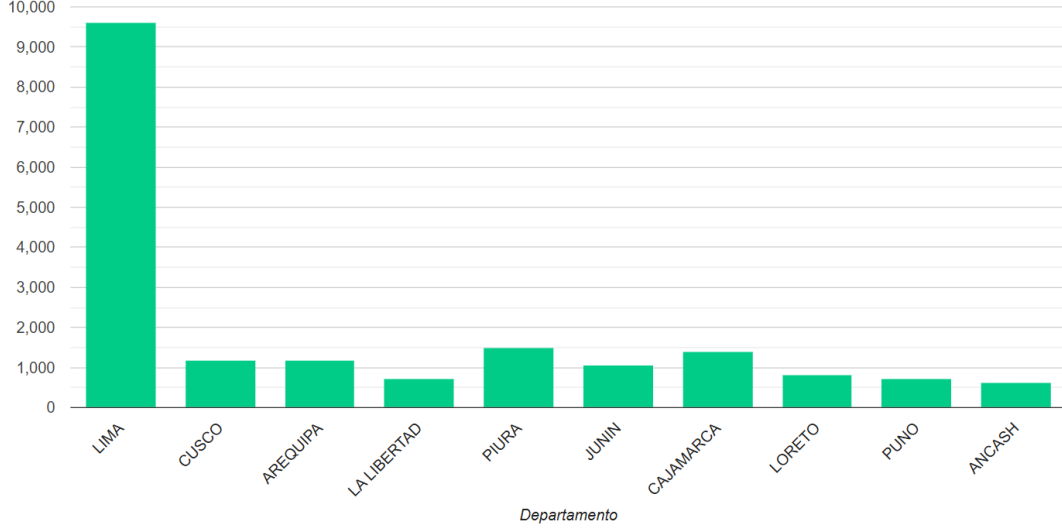


Figure 1: Geographic concentration of facilities across departments

The departmental distribution reveals significant disparities, with Lima containing 45% of all facilities despite housing approximately 29% of the national population, suggesting potential over-concentration in the capital region relative to demographic distribution.

3.3 Geographic Barrier Analysis

3.4 Multi-Scale Buffer Sensitivity Analysis

The sensitivity analysis across four buffer radii (5km, 10km, 15km, 20km) demonstrates how coverage estimates vary with distance thresholds. The 5km WHO-recommended standard, while appropriate for flat terrain with good infrastructure, proves restrictive for Peru’s geographic context. The 10km adapted threshold represents a pragmatic compromise between international standards and local realities, accounting for terrain-imposed travel time increases.

The incremental coverage gains diminish with expanding buffer radii, indicating that populations beyond 10km from existing facilities face substantially greater barriers requiring either new facility construction or alternative service delivery models such as mobile health units or telemedicine.

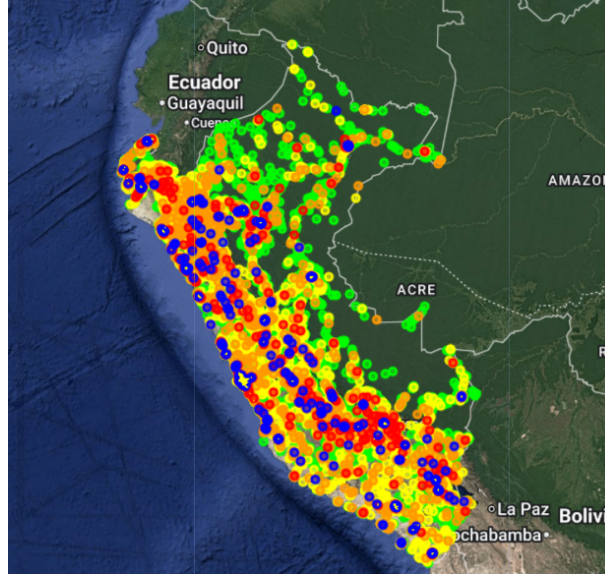


Figure 2: Multi-scale buffer analysis comparing WHO standard versus adapted thresholds

4 Discussion

The results show persistent geographic inequalities in access to healthcare in Peru, limiting progress toward universal health coverage. Even when applying a more realistic 10 km accessibility threshold, substantial gaps remain in Amazonian and high-Andean regions. Healthcare facilities are heavily concentrated in Lima, which holds nearly half of all facilities despite having less than one-third of the national population, indicating inefficiencies in resource distribution typical of many LMICs.

Geographic barriers such as high elevation, steep terrain, and extreme isolation significantly affect both patients and healthcare workers. Facilities located above 3,000 m or far from urban centers face limitations in emergency response, referral systems, and workforce retention. These findings highlight the need for differentiated service models, including telemedicine, incentives for rural healthcare workers, and improved transport infrastructure, particularly for the 14.8

Private healthcare facilities are clustered in economically developed urban zones, reinforcing geographic inequities. While the private sector increases system capacity, its absence in rural areas reflects market failures that necessitate public intervention through regulation, incentives, or public-private partnerships to improve equitable access.

Methodologically, the study demonstrates the value of cloud-based geospatial analysis using Google Earth Engine for national-scale accessibility assessments in resource-constrained settings. However, limitations include reliance on Euclidean distances, lack of road and terrain-adjusted travel times, uniform population assumptions, and the absence of facility-level service differentiation.

From a policy perspective, the findings support targeted infrastructure investment

in underserved areas, expansion of mobile and telehealth services, and more equitable, population- and geography-sensitive resource allocation. Future research should integrate accessibility measures with health outcomes, incorporate transport networks and seasonal variability, and assess access to specialized services to better inform healthcare planning.

5 Conclusions

This study developed and validated a cloud-based Google Earth Engine application for multi-scale geographic accessibility analysis to healthcare facilities in Peru, demonstrating a reproducible methodology applicable to other LMICs facing similar challenges. The analysis of 21,379 facilities revealed substantial geographic disparities, with extreme concentration in Lima department and significant coverage gaps in remote Amazonian and high-Andean regions.

Geographic barriers including high elevation (13.3% of facilities above 3,000m), steep terrain (19.8% on slopes exceeding 20°), and extreme isolation (14.8% over 180 minutes from cities) differentially affect vulnerable rural populations. The pyramidal structure of the healthcare network, with 44.9% Category I-1 basic posts and only 3.3% hospitals, reflects a primary care orientation appropriate for geographically dispersed populations, though gaps in higher-level facility access remain evident.

The adapted 10km accessibility threshold, while more realistic than WHO’s 5km standard for Peru’s terrain, still identifies substantial populations beyond reasonable geographic access to health services. Private sector facilities demonstrate marked clustering in economically developed urban areas (high nighttime radiance zones), creating redundant coverage in cities while leaving rural populations dependent on resource-constrained public facilities. This spatial inequity requires policy interventions to either incentivize private sector rural presence or strengthen public facility capacity in underserved areas.

These findings support targeted policy interventions including: (1) prioritized facility construction in underserved zones identified through spatial gap analysis, (2) enhanced telemedicine infrastructure for isolated facilities exceeding 180 minutes from urban centers, (3) differentiated staffing and resource allocation standards for high-barrier locations, (4) incentive programs for healthcare workers serving in extreme elevation, steep terrain, or isolated contexts, and (5) public-private partnership frameworks encouraging service provision in underserved areas. By providing a free, open-access methodology eliminating traditional barriers to geospatial analysis, this research offers a scalable framework for evidence-based health policy formulation and monitoring of progress toward universal health coverage goals. The application enables health ministries and regional governments to conduct routine accessibility assessments without requiring specialized technical expertise or computational infrastructure, facilitating data-driven resource allocation and infrastructure planning in resource-constrained settings. The demonstrated feasibility of

cloud-based accessibility analysis suggests potential for expansion to other dimensions of health system performance, including temporal accessibility monitoring, service quality integration, and disease-specific access requirements. As satellite data archives continue expanding and computational platforms evolve, such tools will become increasingly powerful for addressing global health equity challenges in geographically complex contexts.

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